KODBTW / VIVA PREPARATION

Frontend application and user experience

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This guide follows the React single-page app from route selection through authentication, theme, data loading and the responsive dashboard. Use the running UI alongside the code when presenting.

Contribution reminder: This is a repository-aligned role guide. Describe work you personally performed; confirm individual contribution details with your team.

Explain the system in one flow

BrowserRouter	**THEN**	**Route + guard**	**THEN**	**Page component**	**THEN**	**API client / Auth context**	**THEN**	**Loading, success, empty or error UI**

Core ideas to explain

App setup

The app mounts BrowserRouter, AuthProvider, ThemeProvider and the route tree. The initial theme is applied before React renders.

Routes

The public landing page and public profile route are accessible without login. Login/register are guest-only; dashboard, analytics, contests, leaderboard, profile and coding profiles are protected.

API client

A shared client adds the Bearer token, applies a 15-second request timeout and translates unauthorized responses into a session event.

Theme

ThemeContext reads the saved preference or system preference, applies light/dark theme attributes and lets the user switch modes.

Experience states

Pages show loading, error and empty states instead of treating missing data as a successful zero. Layout styles adapt across screen widths.

Viva quick preparation

Use these as short answer outlines. Explain the idea in your own words and point to the relevant repository code.

Why use route guards?

They keep protected screens from rendering until the session has been checked. The API still enforces authorization; a frontend guard alone is not security.

How does a restored login work?

The auth provider checks the saved token with `/me`. If validation succeeds it restores the user. If the service is temporarily unreachable, the UI preserves the token and avoids falsely treating a network outage as logout.

How does theme preference persist?

The selected theme is saved in localStorage. On first load, the app can use the system preference, then applies a theme data attribute used by the styles.

What UX states should be shown?

Loading while requests are pending; a helpful error if they fail; an empty state when no records exist; and actual data on success.

Is the client timeout a server timeout?

No. The 15-second setting is the frontend request deadline. Backend, proxy and upstream provider timeouts are separate.

Where to point in the repository

```
01 frontend/src/main.tsx
02 frontend/src/App.tsx
03 frontend/src/contexts/AuthContext.tsx
04 frontend/src/contexts/ThemeContext.tsx
05 frontend/src/api/apiClient.ts
06 frontend/src/pages/DashboardPage.tsx
07 frontend/src/styles/
```

30-second presentation

“The frontend is a React and TypeScript single-page app. React Router selects a page, while auth context restores the current session and protects private screens. The shared API client attaches the token and handles timeout and unauthorized responses. Theme context supports saved light or dark mode. The pages are designed to communicate loading, empty and error states clearly and adapt to smaller screens.”

Be precise A protected route improves navigation but does not replace backend authorization. The public profile route is public by design; do not show private account data in a presentation or screenshot.